

ABM ENGINE

COMPLETE BUSINESS LOGIC BIBLE

AI-Native Account-Based Marketing Platform
KSA Banking & Financial Institutions Ecosystem

FOR: Decimal Technologies
PURPOSE: Complete Business Logic Handoff to Technical Team
DATE: June 10, 2026
VERSION: 2.0 — Ground Zero Build

DOCUMENT INTENT

This document contains ONLY business logic. No code. No technical implementation details.
Every rule, condition, threshold, timing window, decision tree, and edge case is specified.
Your technical developer reads this and builds exactly what the business needs.
If a logic is not in this document, it does not exist in the system.
This is the SINGLE SOURCE OF TRUTH for all ABM engine behavior.

TABLE OF CONTENTS

- Section 1** — FOUNDATIONAL PHILOSOPHY & OPERATING PRINCIPLES
- Section 2** — MASTER ENTITY MODEL — WHAT EXISTS IN THE SYSTEM
- Section 3** — MODULE 1: SIGNAL INTELLIGENCE ENGINE
- Section 4** — MODULE 2: ACCOUNT SCORING ENGINE
- Section 5** — MODULE 3: TIER ASSIGNMENT & ROUTING ENGINE
- Section 6** — MODULE 4: ENRICHMENT ENGINE
- Section 7** — MODULE 5: AI PERSONALIZATION ENGINE (Multi-Agent)
- Section 8** — MODULE 6: QUALITY CONTROL ENGINE
- Section 9** — MODULE 7: OUTREACH EXECUTION ENGINE
- Section 10** — MODULE 8: ENGAGEMENT TRACKING ENGINE
- Section 11** — MODULE 9: HUMAN HANDOFF ENGINE
- Section 12** — MODULE 10: CRM SYNCHRONIZATION ENGINE
- Section 13** — MODULE 11: ANALYTICS & FEEDBACK LOOP ENGINE
- Section 14** — STATE MANAGEMENT — COMPLETE STATE MACHINE
- Section 15** — RETRY, FAILURE & RECOVERY LOGIC
- Section 16** — TIMING MATRIX — EVERY CLOCK IN THE SYSTEM
- Section 17** — KSA-SPECIFIC BUSINESS RULES
- Section 18** — EDGE CASES & EXCEPTION HANDLING
- Section 19** — GLOSSARY OF BUSINESS TERMS

***NOTE:** Each section is self-contained. Your technical developer can implement module by module in the order listed.*

SECTION 1 — FOUNDATIONAL PHILOSOPHY & OPERATING PRINCIPLES

Before any logic is written, the technical team must understand the five non-negotiable operating principles that govern every single decision in the ABM engine. These are not suggestions. They are hard constraints. If any module violates these principles, it is wrong — regardless of whether it technically works.

1.1 — Principle 1: Event-Driven Architecture

Nothing happens without a trigger. The system does NOT run on schedules alone. Every workflow, every score update, every outreach decision, every enrichment request — all are triggered by an **event** (a signal, an engagement, a state change, a timer expiry). The system watches the world and reacts.

***WHY:** Enterprise buying intent is detectable through market events BEFORE prospects enter procurement cycles. The system that detects first, wins.*

What qualifies as a triggering event:

- A new signal is ingested (news, hiring, regulatory, funding)
- An account score crosses a tier threshold (crosses 75, drops below 50)
- An engagement event occurs (email open, reply, LinkedIn accept)
- A timer expires (sequence step delay, freshness window, cooldown period)
- A state transition occurs (account moves from ENRICHING to HOT)
- A human action occurs (manual override, message approval, CRM update)

CRITICAL: The system must NEVER autonomously initiate outreach without a traceable triggering event. Every outbound message must have an audit trail back to the signal or event that caused it.

1.2 — Principle 2: Account-Centric Design

The system is NOT lead-centric. It is **account-centric**. The fundamental unit of work is the ACCOUNT (a bank, fintech, NBFC, digital bank). Contacts are children of accounts. Signals belong to accounts. Scores belong to accounts. Messages are sent on behalf of account strategies.

What this means in practice:

- One account can have multiple contacts — outreach is coordinated ACROSS all contacts of one account
- Scoring happens at the account level, not the contact level
- If Contact A at Bank X replies, the ENTIRE account's outreach pauses — not just Contact A's sequence
- CRM opportunities are created at the account level
- Analytics roll up to account level first, then drill down to contact and channel

1.3 — Principle 3: AI Assists Humans, Never Replaces Them

The AI generates research, personalization, and outreach drafts. Humans make final decisions on enterprise relationships. During MVP, EVERY AI-generated message is reviewed by a human before sending. This is not optional.

AI handles:

- Research and intelligence gathering
- Account scoring and prioritization
- Pain inference and persona analysis
- Message draft generation
- Engagement pattern detection

Humans handle:

- Final message approval (MVP — mandatory)
- Relationship building and trust
- Negotiation and procurement discussions
- Enterprise security and compliance reviews
- Strategic override of AI recommendations

1.4 — Principle 4: Timing Over Volume

The system sends **low volume, high relevance** outreach. This is NOT a mass email tool. Enterprise conversion depends on reaching the right person at the right moment with the right context. One perfectly timed message outperforms 100 generic ones.

Hard constraints:

- Maximum 5 new accounts enter outreach per day during MVP
- Maximum 4 email touches per contact per sequence (spread over 14 days)
- Maximum 3 LinkedIn touches per contact per sequence (spread over 10 days)
- Minimum 72-hour gap between email and LinkedIn first touch to same contact
- Zero outreach to any account without at least one qualifying signal less than 30 days old

1.5 — Principle 5: Reasoning Before Messaging

The AI must NEVER directly generate outreach from raw data. It must follow a structured reasoning chain. Every message has a strategic thought process behind it — not just a template filled with variables.

Mandatory reasoning sequence (no shortcuts):

- **Step 1 — Signal Interpretation:** What happened? Why does it matter to Decimal?
- **Step 2 — Account Research:** What is this organization? What are their strategic priorities?
- **Step 3 — Pain Inference:** Based on signals + research, what operational pains exist?
- **Step 4 — Persona Analysis:** Who is the target contact? What motivates their role?
- **Step 5 — Strategy Decision:** What outreach angle? What value proposition? What ask?
- **Step 6 — Message Generation:** Draft the actual message based on all prior reasoning
- **Step 7 — Quality Validation:** Check for hallucinations, tone, enterprise suitability

CRITICAL: If ANY step is skipped, the output is rejected. The system must enforce this chain — partial reasoning = rejected output.

SECTION 2 — MASTER ENTITY MODEL — WHAT EXISTS IN THE SYSTEM

Every piece of data in the ABM engine belongs to one of the following entities. These are the business objects. Your technical team decides how to store them — but WHAT they store is defined here.

2.1 — ACCOUNT Entity

An account is any organization in the KSA banking and financial institutions ecosystem that Decimal could sell to.

Field	Description	Business Rules
account_name	Legal or commonly known name of the organization	Must be unique. Cannot be blank.
segment	Islamic Bank / Digital Bank / Commercial Bank / BNPL / Embedded Finance / SME Lender / Open Banking / Infrastructure	Exactly one segment. Drives which signal sources matter most.
country	Operating country	MVP: KSA only. Must be 'SA'.
current_tier	HOT / WARM / COLD	Calculated by scoring engine. Cannot be manually set except by admin override.
current_state	WATCHING / ENRICHING / HOT / OUTREACH_ACTIVE / ENGAGED / MEETING_BOOKED / OPPORTUNITY / CUSTOMER / DORMANT	Governed by state machine (Section 14). Transitions are event-driven.
digital_maturity_score	1-10 rating of infrastructure modernization	Set during enrichment. Affects pain inference angles.
open_banking_readiness	1-10 rating of regulatory preparedness	Set during enrichment. Higher = more likely to need Decimal.
procurement_complexity	LOW / MEDIUM / HIGH / EXTREME	Affects sequence length and human involvement level.
relationship_status	NONE / COLD / WARM / ACTIVE / CUSTOMER	Different from tier. This is the EXISTING relationship with Decimal.
assigned_owner	Human sales team member responsible	Assigned when account enters HOT tier. Null for COLD/WARM.
last_signal_date	Timestamp of most recent signal	Used for freshness checks. Account with no signal in 60 days = DORMANT candidate.
last_outreach_date	Timestamp of most recent outbound message	Used for cooldown calculations.
total_score	Current composite score (0-100)	Recalculated on every new signal or engagement event.
created_at	When account was first added	Immutable.
updated_at	Last modification timestamp	Auto-updated on any change.

2.2 — CONTACT Entity

A contact is an individual person at an account who is part of the buying committee or influence chain.

Field	Description	Business Rules
contact_name	Full name	Required.
account_id	Parent account	Every contact MUST belong to exactly one account.
title	Job title	Used for persona classification.
persona_type	CTO / CXO / VP_ENGINEERING / HEAD_DIGITAL / PRODUCT_MANAGER / COMPLIANCE / PROCUREMENT / C_SUITE	Derived from title. Drives message tone and pain angles.
email	Business email	Must be verified before email outreach. Unverified = LinkedIn only.
email_verified	TRUE / FALSE	Set by enrichment verification step.
linkedin_url	LinkedIn profile URL	Required for LinkedIn outreach.
seniority_level	C_LEVEL / VP / DIRECTOR / MANAGER / INDIVIDUAL	Affects outreach formality and strategy.
contact_state	UNCONTACTED / SEQUENCED / OPENED / REPLIED / QUALIFIED / DISQUALIFIED	Governed by engagement events.
preferred_language	EN / AR	Drives message language. Default: EN.
last_contacted_date	Last outbound to this contact	Used for cooldown.
reachability_score	0-100	Composite of email verification + LinkedIn presence + response history.

2.3 — SIGNAL Entity

A signal is any external market event that indicates potential buying intent or strategic relevance.

Field	Description	Business Rules
signal_type	LEADERSHIP_HIRE / DIGITAL_INITIATIVE / REGULATORY / FUNDING / OPEN_BANKING / API_MODERNIZATION / PARTNERSHIP / LICENSING / HIRING_SURGE / PRODUCT_LAUNCH	Exactly one type per signal.
source	GOOGLE_NEWS / SAMA_RSS / LINKEDIN / MANUAL / FUNDING_DB	Where the signal was detected.
account_id	Which account this signal relates to	Must be mapped. Unmapped signals go to a triage queue.
raw_content	Original text/snippet of the signal	Stored verbatim. Never modified after ingestion.
normalized_summary	AI-processed plain-language summary	Max 200 words. Generated during normalization.

Field	Description	Business Rules
signal_strength	P1_CRITICAL / P2_HIGH / P3_MEDIUM / P4_LOW	Assigned by classification rules (see Section 3).
confidence_score	0.0 to 1.0	How confident the system is this signal is real and relevant.
freshness_status	FRESH / AGING / STALE / EXPIRED	Time-based. See freshness rules.
impact_score	0-100	How much this signal affects scoring.
detected_at	When the system found this signal	Immutable.
is_duplicate	TRUE / FALSE	Set by deduplication logic.

CRITICAL: Signals must NEVER be overwritten or deleted. They are append-only. Historical signals are critical for AI memory, analytics, attribution, and learning loops.

2.4 — Additional Core Entities (Summary)

Entity	Purpose	Key Business Rule
ACCOUNT_SCORE	Historical record of every score calculation	New row on every recalculation. Never update in place.
ENRICHMENT	Intelligence gathered about account/contact	Refresh every 30 days for HOT accounts, 90 days for WARM.
AI_GENERATION	Every AI prompt + output + QC result	Must store prompt, output, model, QC score, approval. Critical for auditing.
OUTREACH_SEQUENCE	A planned series of touches to a contact	One active sequence per contact per channel. Never two simultaneous email sequences.
MESSAGE	Every outbound message sent	Must link to sequence, contact, account, AI generation, and channel.
ENGAGEMENT_EVENT	Every inbound signal from prospect	Opens, clicks, replies, LinkedIn accepts, meeting bookings.
OPPORTUNITY	A qualified sales opportunity in CRM	Created only when account reaches MEETING_BOOKED state.

SECTION 3 — MODULE 1: SIGNAL INTELLIGENCE ENGINE

The Signal Intelligence Engine is the system's eyes and ears. It monitors external sources continuously, detects events that indicate buying intent, normalizes them into a structured format, deduplicates them, classifies their priority, and triggers downstream workflows. This is where EVERYTHING begins.

3.1 — Signal Sources & Polling Schedule

Each signal source is polled on a specific schedule. The frequency is based on how fast data changes at that source.

Source	What It Detects	Polling Frequency	WHY This Frequency
Google News API	Market news, leadership changes, M&A, product launches, partnerships	Every 4 hours	News breaks throughout the day. 4-hour windows catch same-day signals without API overuse.
SAMA RSS Feed	Regulatory updates, licensing, open banking mandates, compliance changes	Every 6 hours	Regulatory changes are less frequent but very high impact. 6 hours is sufficient.
LinkedIn (via Heyreach/Phantom)	Job changes, company posts, hiring surges, executive moves	Every 12 hours	LinkedIn data changes slowly. Over-polling risks rate limits and account bans.
Funding Databases (Crunchbase/manual)	Funding rounds, investment announcements	Every 24 hours	Funding rounds are announced at most daily. One daily sweep catches everything.
Manual Input	BD team intelligence, conference notes, relationship insights	On-demand (event-driven)	Humans enter signals when they discover them. No polling needed.

3.2 — Signal Detection Rules

For each source, the system applies specific detection rules to identify relevant signals. Not every news article or LinkedIn post is a signal. The system must filter aggressively.

3.2.1 — Google News Detection Rules

[SIG-DET-001]

INCLUDE only articles that mention a KSA bank/fintech by name AND contain at least one of these intent keywords: 'digital transformation', 'digital banking', 'open banking', 'API', 'core banking', 'lending platform', 'onboarding', 'cloud migration', 'fintech partnership', 'SAMA compliance', 'Islamic finance technology', 'neo bank', 'SME lending', 'origination', 'underwriting platform'.

[SIG-DET-002]

EXCLUDE articles that are: opinion pieces with no factual event, press releases from competitors (Temenos, Backbase, Thought Machine etc. announcing their own wins), articles older than 7 days at time of ingestion,

duplicate URLs already in the system.

[SIG-DET-003]

CONFIDENCE SCORE: If article mentions specific project names or budget figures → confidence 0.9. If article confirms leadership quotes → 0.8. If article is general industry news mentioning the bank → 0.5. If article is tangential → 0.3.

3.2.2 — SAMA RSS Detection Rules

[SIG-DET-010]

INCLUDE any SAMA publication that references: open banking framework, sandbox licenses, compliance deadlines, data residency requirements, new banking licenses, digital-only banking permits, PDPL enforcement.

[SIG-DET-011]

When a SAMA regulatory signal is detected, it applies to ALL accounts in the KSA ecosystem — not just one. The system must fan-out this signal to every active account with a segment affected by the regulation.

[SIG-DET-012]

Regulatory signals are ALWAYS classified as P1_CRITICAL or P2_HIGH. They are never P3 or P4.

3.2.3 — LinkedIn Detection Rules

[SIG-DET-020]

LEADERSHIP_HIRE signal: Detect when a contact at a target account changes to a new role containing: 'CTO', 'CDO', 'Head of Digital', 'VP Technology', 'Chief Information', 'Head of Innovation'. A new leader = new budget, new strategy, new vendor evaluation.

[SIG-DET-021]

HIRING_SURGE signal: If a target account posts 3+ job openings in 14 days with titles containing: 'developer', 'engineer', 'architect', 'platform', 'API', 'cloud', 'digital' — this indicates an active technology build program.

[SIG-DET-022]

COMPANY_POST signal: If a target account's official LinkedIn page posts about: digital transformation milestones, new app launches, customer experience initiatives, technology partnerships — this is a P3 signal (awareness, not action trigger).

3.3 — Signal Normalization Rules

Every signal, regardless of source, must be normalized into a standard structure before it enters the system. Raw data is messy. Normalized data is actionable.

Normalization steps (in exact order):

- **Step 1 — Entity Resolution:** Match the signal to an existing account. Use company name matching with fuzzy tolerance of 85%. If the organization name in the signal matches an existing account at 85%+ confidence, link it. If below 85%, put the signal in the UNMAPPED_SIGNALS queue for human review.
- **Step 2 — Signal Type Classification:** Based on content keywords, assign exactly one signal_type from the allowed list (LEADERSHIP_HIRE, DIGITAL_INITIATIVE, REGULATORY, FUNDING, OPEN_BANKING, API_MODERNIZATION, PARTNERSHIP, LICENSING, HIRING_SURGE, PRODUCT_LAUNCH).
- **Step 3 — Summary Generation:** AI generates a plain-language summary of the signal in under 200 words. This summary must include: WHAT happened, WHO is involved, WHEN it happened, and WHY it matters to Decimal.
- **Step 4 — Priority Classification:** Assign signal_strength based on rules in Section 3.4.
- **Step 5 — Confidence Scoring:** Assign confidence_score from 0.0 to 1.0 based on source reliability and signal specificity.

- **Step 6 — Freshness Tagging:** Set freshness_status to FRESH (detected_at is within 7 days of event date).
- **Step 7 — Impact Scoring:** Calculate impact_score (0-100) based on signal_strength × confidence × freshness weight.

3.4 — Signal Priority Classification Rules

Every signal gets exactly one priority level. This determines how urgently the system acts on it.

Prio rity	Label	Criteria	System Response Time	Examples
P1	CRITICAL	Signal directly indicates immediate vendor evaluation or technology procurement	Score recalculation within 1 hour. Enrichment triggered within 2 hours.	New CTO hired at target bank. SAMA open banking deadline announced. Digital banking RFP published.
P2	HIGH	Signal indicates strategic change that will likely lead to vendor evaluation within 3-6 months	Score recalculation within 4 hours. Enrichment triggered within 24 hours.	Major digital initiative announced. Series B funding round. New digital-only banking license.
P3	MEDIUM	Signal indicates general market activity relevant to Decimal's positioning	Score recalculation within 24 hours. No immediate enrichment.	Company LinkedIn post about digital progress. Industry conference sponsorship. General hiring activity.
P4	LOW	Signal is tangentially relevant — context enrichment only	Score recalculation at next scheduled run. No enrichment.	General industry news mentioning the bank. Market report inclusion. Non-tech partnership.

3.5 — Deduplication Logic

The same event can appear from multiple sources (news picks up what LinkedIn already showed, or the same article appears across multiple news outlets). The system must detect and suppress duplicates.

Deduplication rules:

[SIG-DEDUP-001]

A signal is a DUPLICATE if: (a) same account_id, AND (b) same signal_type, AND (c) detected_at is within the FRESHNESS WINDOW of an existing signal of the same type.

Freshness windows by signal type:

Signal Type	Freshness Window	WHY
LEADERSHIP_HIRE	30 days	A leadership change is reported once. If another article appears within 30 days about the same hire, it's a duplicate.
DIGITAL_INITIATIVE	14 days	A project announcement may get multiple press mentions. 14 days covers the news cycle.
REGULATORY	7 days	SAMA announcements are precise. Duplicates appear within a week.
FUNDING	30 days	Funding rounds get covered by many outlets over weeks.
HIRING_SURGE	14 days	Job postings accumulate. Check over 14-day windows.

Signal Type	Freshness Window	WHY
PRODUCT_LAUNCH	14 days	Product news cycles last about two weeks.
OPEN_BANKING	7 days	Specific regulatory events. Short duplicate window.
API_MODERNIZATION	14 days	Technical announcements.
PARTNERSHIP	14 days	Partnership news.
LICENSING	30 days	License grants are singular events.

[SIG-DEDUP-002]

When a duplicate is detected: Mark the NEW signal as `is_duplicate = TRUE`. Do NOT discard it — store it for audit. Do NOT trigger scoring. Do NOT trigger enrichment.

[SIG-DEDUP-003]

If the duplicate has a HIGHER `confidence_score` than the original, update the original's `confidence_score`. The duplicate still stays marked as duplicate.

3.6 — Signal Freshness Lifecycle

Signals decay over time. A signal from today is more actionable than one from 30 days ago. The system tracks freshness to ensure scoring reflects current reality.

Freshness Status	Age of Signal	Impact on Scoring	System Behavior
FRESH	0-7 days	100% weight	Full scoring impact. Triggers immediate workflows.
AGING	8-21 days	70% weight	Reduced scoring impact. Still considered active context.
STALE	22-45 days	30% weight	Minimal scoring impact. Context only — not an action trigger.
EXPIRED	46+ days	0% weight (excluded)	Excluded from scoring entirely. Remains in DB for historical context and AI memory.

[SIG-FRESH-001]

Freshness status is recalculated every time the account's score is recalculated. It is NOT a scheduled batch job — it is calculated at read time.

[SIG-FRESH-002]

An account with ZERO fresh signals (all signals are STALE or EXPIRED) cannot be in HOT tier. If scoring would place it in HOT, downgrade to WARM with a flag: `'HOT_BLOCKED_NO_FRESH_SIGNAL'`.

3.7 — Signal-to-Scoring Trigger Rules

After a signal passes normalization and deduplication, it must trigger the scoring engine.

[SIG-TRIG-001]

If signal is NOT a duplicate AND `confidence_score >= 0.4` → Trigger account score recalculation immediately for P1/P2 signals, or queue for next scheduled run for P3/P4.

[SIG-TRIG-002]

If signal IS a duplicate → Do NOT trigger scoring.

[SIG-TRIG-003]

If signal cannot be mapped to an account (entity resolution failed) → Place in UNMAPPED_SIGNALS queue. A human reviews within 24 hours. If mapped manually, trigger scoring at that point.

SECTION 4 — MODULE 2: ACCOUNT SCORING ENGINE

The Account Scoring Engine is the brain that decides which accounts deserve attention. It calculates a composite score (0-100) for every account based on four weighted dimensions. The score determines the account's tier (HOT / WARM / COLD) which in turn determines what workflows activate.

4.1 — Scoring Dimensions & Weights

Dimension	Weight	What It Measures	WHY This Weight
Signal Strength	35%	Quality, recency, and volume of buying intent signals	Signals are the primary driver. No signal = no reason to reach out. Strongest predictor of timing.
Regulatory Pressure	30%	Urgency created by SAMA mandates, compliance deadlines, licensing requirements	In KSA, regulatory pressure is the #1 external force driving bank technology spend. SAMA says jump, banks ask how high.
Persona Reachability	20%	Whether we can actually reach decision-makers at this account	A perfectly timed opportunity means nothing if we cannot contact anyone. Reachability gates execution.
Relationship Warmth	15%	Existing relationship strength between Decimal and this account	Warm relationships convert 3-5x better than cold outreach. A warm intro to a hot account is the ideal scenario.

4.2 — Dimension 1: Signal Strength Score (0-100)

Calculated by aggregating all non-duplicate, non-expired signals for the account.

Step 1 — Score each individual signal

Signal Priority	Base Points	Freshness Multiplier	Max Points Per Signal
P1_CRITICAL	40 points	FRESH: ×1.0 AGING: ×0.7 STALE: ×0.3	40
P2_HIGH	25 points	FRESH: ×1.0 AGING: ×0.7 STALE: ×0.3	25
P3_MEDIUM	12 points	FRESH: ×1.0 AGING: ×0.7 STALE: ×0.3	12
P4_LOW	5 points	FRESH: ×1.0 AGING: ×0.7 STALE: ×0.3	5

Step 2 — Aggregate and cap

[SCORE-SIG-001]

Sum all individual signal scores for the account. Apply confidence_score as a multiplier to each signal before summing (signal_points × freshness_multiplier × confidence_score).

[SCORE-SIG-002]

Cap the total at 100. If sum exceeds 100, the Signal Strength Score = 100.

[SCORE-SIG-003]

If account has zero non-expired signals, Signal Strength Score = 0.

[SCORE-SIG-004]

BONUS: If account has 3+ FRESH P1/P2 signals simultaneously, add a 15-point surge bonus (before capping at 100). WHY: Multiple simultaneous high-intent signals indicate active transformation — this is the ideal target.

4.3 — Dimension 2: Regulatory Pressure Score (0-100)

Measures how much regulatory urgency affects this specific account.

[SCORE-REG-001]

Start with base score of 0.

[SCORE-REG-002]

For each active SAMA regulation/mandate that applies to this account's segment: +25 points per regulation with a deadline within 6 months. +15 points per regulation with a deadline within 12 months. +5 points per regulation with no specific deadline but confirmed applicability.

[SCORE-REG-003]

If the account's open_banking_readiness score is LOW (1-3) AND there is an active open banking mandate → add +20 penalty urgency points. WHY: Low readiness + high regulatory pressure = maximum pain = maximum Decimal relevance.

[SCORE-REG-004]

Cap at 100.

[SCORE-REG-005]

If no active regulations apply to this account's segment → Regulatory Pressure Score = 0.

4.4 — Dimension 3: Persona Reachability Score (0-100)

Measures whether we can actually reach decision-makers and influencers at this account.

[SCORE-REACH-001]

For each contact in the buying committee: Verified email + LinkedIn = 30 points. Verified email only = 20 points. LinkedIn only = 10 points. Neither = 0 points.

[SCORE-REACH-002]

Take the average across all contacts at the account. If no contacts → Reachability = 0.

[SCORE-REACH-003]

BONUS: If at least one C-level contact has verified email → +20 bonus points. WHY: C-suite email access is rare and extremely valuable in KSA enterprise sales.

[SCORE-REACH-004]

BONUS: If 3+ contacts across different departments are reachable → +10 points. WHY: Multi-threaded outreach dramatically increases enterprise conversion.

[SCORE-REACH-005]

Cap at 100.

4.5 — Dimension 4: Relationship Warmth Score (0-100)

Measures existing relationship between Decimal and this account.

Relationship Status	Base Score	Modifiers
NONE — No prior interaction	0	Pure cold outreach.
COLD — Some awareness, no active dialogue	15	+5 if Decimal content was shared. +5 if met at conference.
WARM — Active introduction exists through connector	40	+10 if connector is a trusted partner (ITC, Backbase, Tarabut, Geidea, KPMG, Mastercard). +5 if connector has confirmed willingness to intro.
ACTIVE — Ongoing dialogue or past engagement	70	+10 if previous meeting happened. +10 if CRM has positive notes.
CUSTOMER — Existing Decimal client	90	+10 if upsell opportunity identified.

[SCORE-REL-001]

Cap at 100.

4.6 — Final Score Calculation

SCORING FORMULA

$$\text{TOTAL_SCORE} = (\text{Signal_Strength} \times 0.35) + (\text{Regulatory_Pressure} \times 0.30) + (\text{Persona_Reachability} \times 0.20) + (\text{Relationship_Warmth} \times 0.15)$$

Result: A number between 0 and 100, rounded to nearest integer.

[SCORE-CALC-001]

Every score calculation must store a complete record: account_id, total_score, each dimension score, each dimension weight, timestamp, triggering_event (which signal or event caused this recalculation).

[SCORE-CALC-002]

NEVER overwrite previous scores. Each calculation is a new historical record. This is critical for analytics and trend analysis.

[SCORE-CALC-003]

Score must be recalculated when: (a) new non-duplicate signal ingested for this account, (b) engagement event received, (c) enrichment data updated, (d) relationship status changed, (e) scheduled monthly recalibration.

4.7 — Weight Recalibration Rules

Weights are not permanently fixed. They must be recalibrated based on actual conversion data.

[SCORE-RECAL-001]

After 90 days of operation with at least 10 meetings booked, analyze which dimension had the highest correlation with successful meetings. Adjust weights by $\pm 5\%$ increments. Never change more than 5% per dimension per recalibration cycle.

[SCORE-RECAL-002]

Signal Strength weight must NEVER drop below 25%. It is always the primary driver.

[SCORE-RECAL-003]

All weight changes must be logged with: old_weights, new_weights, reason, data_supporting_change, approved_by.

SECTION 5 — MODULE 3: TIER ASSIGNMENT & ROUTING ENGINE

After scoring, each account is assigned a tier. The tier determines what happens next. This is the routing layer — the traffic cop of the entire system.

5.1 — Tier Thresholds

Tier	Score Range	Meaning	What Happens Next
HOT	75-100	High-intent account with active buying signals, reachable contacts, and/or regulatory urgency	Immediate enrichment → AI personalization → Outreach sequence launch
WARM	50-74	Moderately interesting account with some signals but incomplete picture	Add to nurture sequence. Monitor for signal upgrades. Periodic re-enrichment.
COLD	0-49	Low current relevance. Insufficient signals or unreachable.	Monitor only. No outreach. No enrichment. Re-score on new signals.

5.2 — Tier Transition Rules

Upgrading tiers:

[TIER-UP-001]

COLD → WARM: Score crosses 50 on recalculation. Action: Log tier change. Add to nurture monitoring. Schedule enrichment check within 48 hours.

[TIER-UP-002]

WARM → HOT: Score crosses 75 on recalculation. Action: Log tier change. Trigger immediate enrichment workflow. Change account_state to ENRICHING. Assign account_owner if not already assigned.

[TIER-UP-003]

COLD → HOT (skip WARM): Score crosses 75 on first calculation or on a single P1 signal. Action: Same as TIER-UP-002. This happens when a major event hits a previously unknown account.

Downgrading tiers:

[TIER-DOWN-001]

HOT → WARM: Score drops below 75 on recalculation AND account is NOT in OUTREACH_ACTIVE or ENGAGED state. Action: Log tier change. Pause any pending enrichment. Do NOT cancel active outreach sequences — let them complete.

[TIER-DOWN-002]

WARM → COLD: Score drops below 50 AND no engagement events in last 30 days. Action: Log tier change. Remove from nurture sequence.

[TIER-DOWN-003]

HOT → WARM downgrade is BLOCKED if: account_state is OUTREACH_ACTIVE, ENGAGED, or MEETING_BOOKED. WHY: Once we're actively engaging, we don't pull back based on score fluctuations. Human judgment takes over.

Special overrides:

[TIER-OVR-001]

Manual admin override: A human can force-set any account to any tier with a written reason. The reason must be logged. Manual overrides expire after 30 days — then the scoring engine takes over again.

[TIER-OVR-002]

Fresh-signal gate: An account CANNOT be HOT if it has zero FRESH signals (all signals older than 7 days). Even if the composite score is 75+, the account is capped at WARM with flag: 'HOT_BLOCKED_NO_FRESH_SIGNAL'.

5.3 — Routing Logic After Tier Assignment

What happens immediately after a tier is assigned or changed:

IF New tier = HOT AND previous tier was NOT HOT

THEN → Trigger Enrichment Engine (Section 6)

THEN → Change account_state to ENRICHING

THEN → Assign account_owner from sales team rotation

THEN → Send Slack notification: 'Account [NAME] promoted to HOT – Score: [X]'

IF New tier = WARM AND previous tier was COLD

THEN → Add account to nurture monitoring list

THEN → Schedule enrichment check in 48 hours

THEN → Log: 'Account [NAME] moved to WARM – Score: [X]'

IF New tier = COLD AND previous tier was WARM/HOT

THEN → Check if outreach is active – if YES, do NOT change tier (TIER-DOWN-003)

THEN → If no active outreach, remove from nurture

THEN → Log: 'Account [NAME] downgraded to COLD – Score: [X]'

SECTION 6 — MODULE 4: ENRICHMENT ENGINE

The Enrichment Engine builds deep intelligence profiles for accounts that have been promoted to HOT. It gathers company data, identifies the buying committee, verifies contact information, analyzes strategic priorities, and generates AI account summaries. The goal: arm the AI personalization engine with enough context to generate highly relevant outreach.

6.1 — Enrichment Trigger Rules

[ENR-TRIG-001]

Enrichment is triggered when: (a) Account moves to HOT tier for the first time, (b) Account has been HOT for 30+ days and enrichment data is older than 30 days (auto-refresh), (c) Manual enrichment request by account_owner.

[ENR-TRIG-002]

Enrichment is NOT triggered for WARM or COLD accounts (exception: 48-hour enrichment check for newly WARM accounts is a LIGHT enrichment — company data only, no contact discovery).

6.2 — Enrichment Workflow Steps (Exact Order)

Step 1 — Company Data Enrichment

- Fetch: Company size, headquarters, founding year, revenue range, technology stack, recent news, key initiatives, competitive landscape
- Sources: Clearbit, Clay, manual research
- Timeout: If enrichment API does not respond within 30 seconds, retry once after 60 seconds. If second attempt fails, log failure and proceed with available data.

Step 2 — Buying Committee Discovery

- Identify all contacts at the account who hold decision-making or influencing roles
- Minimum contacts to discover: 3 per HOT account. Ideal: 5-7 spanning CTO/CDO, Head of Digital, VP Engineering, Product Lead, Compliance, Procurement.
- Source: Apollo for contact discovery. LinkedIn for validation.
- Rule: If fewer than 3 contacts can be found, flag account as 'LOW_COMMITTEE_COVERAGE' — this reduces Reachability score on next recalculation.

Step 3 — Email Verification

- Every discovered email must be verified before it's used for outreach
- Verification levels: VERIFIED (confirmed deliverable), RISKY (catch-all or unverifiable), INVALID (bounced)
- Rule: Only VERIFIED emails are used for email outreach. RISKY emails get a LinkedIn-only flag. INVALID emails are marked and excluded.
- Timeout: If verification API does not respond in 15 seconds, mark as UNVERIFIED and retry in next enrichment cycle.

Step 4 — Hiring Trend Analysis

- Check if the account has active job postings related to technology/digital
- If 5+ relevant postings are active, generate a HIRING_SURGE signal (if not already detected)
- This creates a feedback loop: enrichment can generate new signals that boost the score further

Step 5 — Strategic Initiative Analysis

- AI analyzes all gathered data to identify the account's top 3 strategic priorities
- Output: A ranked list of 3 strategic initiatives (e.g., 'Digital lending platform', 'Open banking compliance', 'Customer onboarding modernization')
- These priorities are stored and used by the AI Personalization Engine to select outreach angles

Step 6 — AI Account Summary Generation

- AI generates a comprehensive account summary (500-800 words) covering: organization overview, strategic priorities, technology landscape, regulatory pressures, Decimal's relevance, recommended outreach angles
- This summary is stored as the primary context for all future AI reasoning about this account

Step 7 — State Transition

- After enrichment completes successfully: Change account_state from ENRICHING → HOT
- Trigger the AI Personalization Engine (Section 7)
- Log: 'Enrichment complete for [ACCOUNT]. Contacts found: [N]. Verified emails: [N]. Top priority: [X].'

6.3 — Enrichment Refresh Rules

[ENR-REF-001]

HOT accounts: Full re-enrichment every 30 days.

[ENR-REF-002]

WARM accounts: Light enrichment (company data only) every 90 days.

[ENR-REF-003]

COLD accounts: No enrichment.

[ENR-REF-004]

If an account owner manually triggers enrichment, it runs regardless of schedule.

SECTION 7 — MODULE 5: AI PERSONALIZATION ENGINE (Multi-Agent)

This is the most sophisticated module in the system. It uses a multi-agent AI architecture where specialized AI agents handle different aspects of reasoning, and their outputs chain together to produce highly personalized outreach. This is NOT a template engine. Every message is strategically reasoned.

7.1 — Why Multi-Agent Architecture

SINGLE-PROMPT AI FAILS BECAUSE:

- Reasoning becomes shallow when one prompt tries to do everything
- Outputs become generic because context gets diluted
- Hallucination risk increases with prompt complexity
- No quality gates — garbage in, garbage out with no checkpoint
- Cannot specialize — one prompt cannot be an expert researcher AND psychologist AND copywriter

7.2 — The 7 AI Agents (Exact Sequence)

Agent 1 — Signal Analyst

- **Input:** Raw signal data, normalized summary, signal_type, signal_strength
- **Task:** Interpret what this signal means for the account's technology trajectory. WHY is this signal significant? What change does it represent? What downstream effects will it have?
- **Output:** A 100-200 word strategic interpretation
- **Constraint:** Must ONLY use facts from the signal. Must NOT invent any details. Must state confidence level.

Agent 2 — Account Research Agent

- **Input:** Enrichment data, AI account summary, strategic initiatives, technology stack
- **Task:** Build a comprehensive understanding of this organization. What are they trying to achieve? Where are they in their digital journey? What's their competitive position?
- **Output:** Structured account intelligence brief (300-500 words)
- **Constraint:** Must cite sources for every claim. No invented information. Must flag gaps in knowledge.

Agent 3 — Persona Psychology Agent

- **Input:** Contact's title, seniority level, persona_type, department
- **Task:** Analyze what motivates this specific persona. What are their KPIs? What keeps them up at night? What language resonates with them? What do they care about most?
- **Output:** Persona profile (150-250 words) including: primary motivations, decision criteria, communication preferences, likely objections
- **Constraint:** Must be based on role analysis, not personal assumptions. A CTO cares about architecture and scalability. A compliance head cares about regulatory risk. A CDO cares about customer experience metrics.

Agent 4 — Pain Inference Agent

- **Input:** Signal interpretation + Account intelligence + Persona profile
- **Task:** Infer the 2-3 most likely operational pains this persona is facing RIGHT NOW, given the signal context and account situation
- **Output:** Ranked list of 3 inferred pains with confidence levels and reasoning
- **Constraint:** Pain must be INFERRED from data, not assumed generically. 'Probably has manual processes' is too generic. 'Their 2/5 App Store rating suggests onboarding friction that a no-code origination layer could fix' is properly inferred.

Agent 5 — Strategy Agent

- **Input:** All outputs from Agents 1-4
- **Task:** Decide the outreach strategy. What angle do we lead with? What Decimal capability do we highlight? What is the call-to-action? What tone? How direct?
- **Output:** Outreach strategy brief including: lead angle, value proposition, proof point to use, call-to-action, tone guidance (formal/consultative/peer-to-peer), channel recommendation (email-first or LinkedIn-first)
- **Constraint:** Must select ONE primary angle. Not a buffet of options. One clear, decisive strategy.

Agent 6 — Message Generation Agent

- **Input:** Complete strategy brief from Agent 5 + all prior context
- **Task:** Write the actual outreach message (email subject + body, or LinkedIn message)
- **Output:** Complete message ready for human review
- **Constraints (HARD RULES — message must satisfy ALL):**
 - Maximum 150 words for email body. Maximum 80 words for LinkedIn message.
 - Must reference the specific signal or trigger — no generic openers.
 - Must include exactly ONE specific Decimal capability or proof point.
 - Must end with a low-friction CTA (not 'buy our product' — rather 'would a 15-minute call make sense?').
 - Must NOT use: 'I hope this email finds you well', 'Just reaching out', 'I wanted to touch base', 'Synergy', 'Leverage', 'Circle back'.
 - Must NOT claim false familiarity ('As a fellow...') unless relationship_status is WARM or ACTIVE.
 - Must match the persona's seniority: C-suite gets strategic framing. Directors get operational framing. Managers get tactical framing.

Agent 7 — Quality Control Agent

- **Input:** Generated message + all prior reasoning chain
- **Task:** Validate the message against quality, accuracy, and enterprise suitability criteria
- **Output:** QC verdict: APPROVED / REJECTED with specific reasons
- **Full QC validation detailed in Section 8**

7.3 — Language Rules

[AI-LANG-001]

Default outreach language: English.

[AI-LANG-002]

If contact's preferred_language is AR → Generate in Arabic. The Arabic message must be reviewed by an Arabic-fluent human before sending.

[AI-LANG-003]

LinkedIn messages should match the language of the contact's LinkedIn profile activity.

[AI-LANG-004]

NEVER mix languages in a single message. Either full English or full Arabic.

7.4 — AI Model & Temperature Rules

[AI-MODEL-001]

Primary model: Claude Sonnet (latest available version). Fallback: GPT-4o.

[AI-MODEL-002]

Temperature settings: Research agents (1-4): temperature 0.3 (factual, grounded). Strategy agent (5): temperature 0.5 (some creativity). Message generation (6): temperature 0.4 (creative but controlled). QC agent (7): temperature 0.1 (strict, binary).

[AI-MODEL-003]

If primary model fails (timeout, error), retry ONCE with same model. If second attempt fails, try fallback model. If fallback also fails, queue for manual handling.

SECTION 8 — MODULE 6: QUALITY CONTROL ENGINE

The QC Engine is the gatekeeper. No message leaves the system without passing QC. During MVP, QC is a TWO-LAYER process: AI QC Agent validates first, then a human reviews the AI-approved message before it's actually sent.

8.1 — AI QC Validation Checks (All Must Pass)

Check ID	Check Name	Pass Criteria	Failure Action
QC-001	Hallucination Check	Every factual claim in the message can be traced to enrichment data or signal data. No invented facts.	REJECT. Log: 'Hallucination detected: [specific claim]'. Re-queue for Agent 6 regeneration with explicit constraint: 'Do not mention [hallucinated fact]'.
QC-002	Tone Check	Message tone matches enterprise B2B standards. No casual slang, no aggressive sales language, no desperation.	REJECT. Log: 'Tone violation: [specific phrase]'. Re-queue with tone guidance.
QC-003	Length Check	Email body $\leq$ 150 words. LinkedIn message $\leq$ 80 words. Subject line $\leq$ 60 characters.	REJECT. Log: 'Length exceeded: [actual] vs [max]'. Re-queue with strict length constraint.
QC-004	CTA Check	Message contains exactly ONE clear, low-friction call-to-action.	REJECT if 0 CTAs or 2+ CTAs. Log specific issue. Re-queue.
QC-005	Personalization Check	Message references at least ONE specific detail about the account or contact (signal, initiative, role-specific concern).	REJECT if generic. Log: 'No personalization detected'. Re-queue with specific context to include.
QC-006	Banned Phrase Check	Message does NOT contain any banned phrases (listed in Section 7.2 Agent 6 constraints).	REJECT. Log: 'Banned phrase found: [phrase]'. Auto-remove phrase and re-validate.
QC-007	Compliance Check	Message does not make false claims about Decimal, does not guarantee outcomes, does not reference confidential information about other clients.	REJECT. Log: 'Compliance violation: [detail]'. Escalate to human review — do not auto-regenerate.

8.2 — Regeneration Limits

[QC-REGEN-001]

Maximum 3 regeneration attempts per message. If a message fails QC 3 times, it is flagged for MANUAL_CREATION — a human writes it from scratch using the AI's reasoning chain as reference.

[QC-REGEN-002]

Each regeneration attempt must include the specific QC failure reason as an additional constraint. The AI must not repeat the same mistake.

[QC-REGEN-003]

Log every regeneration: attempt_number, failure_reason, regenerated_output, new_qc_result. This data is critical for prompt optimization.

8.3 — Human Review Layer (MVP Mandatory)

[QC-HUMAN-001]

During MVP (first 90 days minimum): Every AI-approved message goes to a human reviewer before sending. No exceptions.

[QC-HUMAN-002]

Human reviewer can: APPROVE (message sends), EDIT (modify and approve), REJECT (message does not send, logged with reason).

[QC-HUMAN-003]

Human review SLA: Messages must be reviewed within 4 hours during business hours (9am-6pm KSA time, Sunday-Thursday). Messages queued outside business hours are reviewed at start of next business day.

[QC-HUMAN-004]

After 90 days, if AI QC approval rate exceeds 85% AND human override rate is below 10%, the human review can be made optional for P3/P4 signals. P1/P2 signals ALWAYS get human review, even post-MVP.

SECTION 9 — MODULE 7: OUTREACH EXECUTION ENGINE

The Outreach Execution Engine sends approved messages through email and LinkedIn channels. It manages sequences (multi-touch campaigns), coordinates timing across channels, handles delivery, and detects replies.

9.1 — Channel Strategy

[OUT-CHAN-001]

Primary channel: Email (for contacts with verified emails). Secondary channel: LinkedIn (for all contacts with LinkedIn URLs).

[OUT-CHAN-002]

Dual-channel coordination: If a contact has both verified email AND LinkedIn, run email sequence FIRST. Start LinkedIn sequence 72 hours after first email. WHY: Email is more professional for initial enterprise outreach. LinkedIn adds social presence after email opens the door.

[OUT-CHAN-003]

LinkedIn-only: If contact has no verified email, run LinkedIn sequence as primary and only channel.

9.2 — Email Sequence Design

Touch	Timing	Objective	Content Strategy	Send Window
Touch 1	Day 1 (immediate after approval)	Signal-triggered opener	Reference the specific signal. Show Decimal's relevance. Ask for 15-min conversation.	Tue-Thu, 9am-11am KSA time
Touch 2	Day 4 (+3 days after Touch 1)	Value-led insight	Share a specific insight about their initiative. Link to a relevant case study or data point. No hard sell.	Sun-Thu, 9am-2pm KSA time
Touch 3	Day 8 (+4 days after Touch 2)	Social proof / comparable	Reference a similar bank that achieved results with Decimal. Keep it brief and specific.	Sun-Thu, 9am-2pm KSA time
Touch 4	Day 14 (+6 days after Touch 3)	Low-friction close	Final attempt. Acknowledge they're busy. Offer something ultra-easy: '2-minute Loom video showing how we solved X for [comparable bank]?'	Tue-Thu, 9am-11am KSA time

9.3 — LinkedIn Sequence Design

Step	Timing	Objective	Content Strategy
Step 1	Day 1 (or Day 4 if dual-channel)	Connection request	Personalized note referencing shared context (signal, industry, mutual connection). Max 300 characters.
Step 2	+2 days after connection accepted	Value DM	Share an insight relevant to their role. Not a pitch. A genuine observation about their strategic situation.
Step 3	+7 days after Step 2	Contextual follow-up	Reference something they posted or a new signal. Ask for a conversation. Final LinkedIn touch.

9.4 — Sequence Execution Rules

[OUT-SEQ-001]

NEVER run two active email sequences for the same contact simultaneously. One email sequence at a time per contact.

[OUT-SEQ-002]

NEVER run two active LinkedIn sequences for the same contact simultaneously.

[OUT-SEQ-003]

If a reply is detected on ANY channel → IMMEDIATELY PAUSE ALL sequences for ENTIRE ACCOUNT (all contacts). WHY: Account-centric design. A reply on one contact means the account is engaged — hitting other contacts simultaneously looks uncoordinated.

[OUT-SEQ-004]

If an email bounces → Mark contact email as INVALID. Do NOT retry. Switch to LinkedIn-only for this contact.

[OUT-SEQ-005]

If LinkedIn connection request is ignored after 7 days → DO NOT resend. Mark as 'connection_pending'. Try again only if a new P1 signal arrives (minimum 30 days gap).

[OUT-SEQ-006]

Cooldown between completed sequence and new sequence for same contact: 45 days minimum. WHY: Enterprise buyers find repeated campaigns annoying. Wait for a new signal or context change.

[OUT-SEQ-007]

Maximum outreach velocity: 15 emails per day across all contacts (not per contact — total system output per day during MVP). WHY: Deliverability. New domains need low-volume warmup.

9.5 — Deliverability Rules

[OUT-DELIV-001]

All outreach emails must be sent from dedicated outreach domains (NOT decimal.tech primary domain). WHY: Protect primary domain reputation.

[OUT-DELIV-002]

Domain warmup: New domains send max 5 emails/day for week 1, 10/day for week 2, 15/day for week 3, then stabilize at 20/day max.

[OUT-DELIV-003]

Mandatory authentication: SPF, DKIM, DMARC must be configured BEFORE any email is sent. If not configured → block all email sending.

[OUT-DELIV-004]

Bounce rate monitoring: If bounce rate exceeds 5% in any 7-day window → PAUSE all email sending. Investigate contact quality. Resume only after root cause is fixed.

SECTION 10 — MODULE 8: ENGAGEMENT TRACKING ENGINE

The Engagement Tracking Engine monitors all prospect interactions with the outreach the system sends. Every open, click, reply, LinkedIn accept, and meeting booking is captured, classified, and used to trigger downstream workflows.

10.1 — Engagement Events & Classification

Event	Source	Intent Level	What It Triggers
Email Open	Email tool webhook	LOW — awareness signal only	Update engagement score. Log event. Do NOT trigger handoff.
Email Click	Email tool webhook	MEDIUM — active interest	Update engagement score. Log event. Flag contact as 'interested'.
Email Reply (positive)	Email tool + sentiment analysis	HIGH — active engagement	PAUSE all sequences for account. Trigger human handoff. Create CRM task.
Email Reply (negative/unsubscribe)	Email tool + sentiment analysis	DISQUALIFICATION	STOP all sequences for this contact permanently. Log. Do NOT outreach again unless manual override.
Email Reply (neutral/auto-reply)	Email tool + sentiment analysis	NONE — informational	Log only. Continue sequence as planned.
LinkedIn Connection Accepted	LinkedIn tool webhook	MEDIUM — social engagement	Trigger LinkedIn DM sequence (Step 2). Update engagement score.
LinkedIn Connection Ignored (7+ days)	Timer	LOW — no interest	Mark as 'connection_pending'. Stop LinkedIn sequence for this cycle.
LinkedIn Message Reply	LinkedIn tool webhook	HIGH — active engagement	PAUSE all sequences for account. Trigger human handoff.
Meeting Booked	Calendar tool / CRM	CRITICAL — conversion event	Change account_state to MEETING_BOOKED. Trigger AI meeting brief. Notify sales team.

10.2 — Sentiment Classification Rules

When a reply is received, the system must classify its sentiment to determine the correct response.

[ENG-SENT-001]

POSITIVE reply indicators: mentions 'interested', 'let's talk', 'send me more', 'schedule', 'meeting', 'call', 'demo', 'tell me more', asks a question about Decimal's capabilities, forwards to a colleague.

[ENG-SENT-002]

NEGATIVE reply indicators: 'not interested', 'unsubscribe', 'remove me', 'stop emailing', 'we already have a vendor', 'no budget', 'bad timing', explicit rejection.

[ENG-SENT-003]

NEUTRAL reply indicators: out-of-office auto-reply, 'let me check internally', 'not the right person — try [name]', informational response without commitment.

[ENG-SENT-004]

REFERRAL reply: Contact says 'you should talk to [name]' → This is POSITIVE. Capture the referred contact name. Create a new contact record. Generate a personalized outreach for the referral mentioning the introduction.

[ENG-SENT-005]

AI classifies sentiment first. If confidence is below 0.7, escalate to human classification. Human classification overrides AI.

10.3 — Engagement Scoring

Engagement score is accumulated per contact and rolled up to account level.

Event	Points	Notes
Email Open	+2 per open (max 3 opens counted)	After 3 opens, additional opens add 0 points. Prevents score inflation from email preview panes.
Email Click	+5 per click	Each unique link click counted once.
Email Positive Reply	+25	Only counted once per sequence.
LinkedIn Accept	+10	Once per contact.
LinkedIn Reply	+25	Only counted once.
Meeting Booked	+50	The ultimate conversion event.
Email Negative Reply	-30	Penalizes the contact and reduces account engagement score.

[ENG-SCORE-001]

Account engagement score = sum of all contact engagement scores at that account. This is used in the feedback loop to refine scoring weights and signal quality assessment.

SECTION 11 — MODULE 9: HUMAN HANDOFF ENGINE

When a prospect shows positive engagement, the system transitions from AI-automated outreach to human-led enterprise sales. This handoff must be seamless — the human must receive complete context so the prospect never feels like they're starting over.

11.1 — Handoff Trigger Conditions

Trigger	Priority	Assigned To	Response SLA
Positive email reply	URGENT	Account owner (SDR)	Respond within 2 hours during business hours
Meeting request	CRITICAL	Account owner (AE)	Confirm within 1 hour
Pricing question	HIGH	Account Executive	Respond within 4 hours
Security / compliance question	HIGH	Technical team + AE	Respond within 24 hours (needs research)
Referral to colleague	MEDIUM	Account owner (SDR)	New outreach within 24 hours
LinkedIn positive reply	URGENT	Account owner	Respond within 2 hours

11.2 — Handoff Package (What Human Receives)

When a handoff is triggered, the system generates a complete context package:

- **Account Summary:** AI-generated brief of the organization, strategic priorities, and digital maturity
- **Signal Timeline:** All signals for this account in reverse chronological order with interpretations
- **Engagement History:** Every message sent to every contact at this account, with opens/clicks/replies
- **Reply Content:** The actual reply text that triggered the handoff
- **Persona Profile:** The AI-generated persona analysis for the replying contact
- **Pain Inference:** Top 3 inferred pains from the AI reasoning chain
- **Suggested Discussion Angles:** AI-generated talking points for the first human conversation
- **Likely Objections:** AI-predicted objections with suggested responses
- **Comparable Wins:** Relevant Decimal case studies or client wins similar to this account's situation

11.3 — Handoff Workflow Steps

[HAND-001]

Step 1: PAUSE all automated sequences for the entire account (all contacts, all channels). This is immediate and automatic.

[HAND-002]

Step 2: Change account_state to ENGAGED.

[HAND-003]

Step 3: Generate AI meeting preparation brief (handoff package above).

[HAND-004]

Step 4: Send Slack notification to account_owner with: account name, contact who replied, reply text, link to full brief.

[HAND-005]

Step 5: Create CRM task in HubSpot: 'Follow up with [CONTACT] at [ACCOUNT] — positive reply received [DATETIME]'.

[HAND-006]

Step 6: Start a 48-hour timer. If account_owner does not acknowledge the handoff within 48 hours, escalate to sales manager.

SECTION 12 — MODULE 10: CRM SYNCHRONIZATION ENGINE

The CRM (HubSpot) is the central system of record for sales teams. The ABM engine must synchronize data bidirectionally — pushing intelligence into HubSpot and pulling sales updates back.

12.1 — ABM → HubSpot (Push)

Data	When	Where in HubSpot
Account tier (HOT/WARM/COLD)	On every tier change	Company property: 'abm_tier'
Account score	On every score recalculation	Company property: 'abm_score'
Contact records	On enrichment completion	Contact records linked to company
Engagement events	On every engagement	Activity timeline on contact
Opportunity creation	When account reaches MEETING_BOOKED	Deal in pipeline
AI meeting brief	On handoff trigger	Note attached to deal
Outreach history	After each message sent	Activity on contact record

12.2 — HubSpot → ABM (Pull)

Data	When	Effect on ABM Engine
Deal stage changes	Polled every 30 minutes	Updates account_state to OPPORTUNITY if deal created
Meeting outcomes	Polled every 30 minutes	Feeds back into AI learning: was the meeting successful?
Manual contact additions	Polled every 60 minutes	New contacts are added to the enrichment queue
Deal closed-won	On event	Account_state → CUSTOMER. All outreach permanently stopped.
Deal closed-lost	On event	Account_state → DORMANT. 90-day cooldown before any re-engagement.

[CRM-SYNC-001]

HubSpot is ALWAYS the source of truth for deal stage and pipeline value. The ABM engine does NOT override HubSpot deal data.

[CRM-SYNC-002]

If CRM sync fails, log the failure and retry with exponential backoff: 1 min, 5 min, 15 min, 60 min, then dead-letter queue for manual handling.

SECTION 13 — MODULE 11: ANALYTICS & FEEDBACK LOOP ENGINE

The feedback loop is what makes the system intelligent over time. Without it, the ABM engine is static. With it, the engine learns what works and self-optimizes.

13.1 — Core KPIs

KPI	Definition	Target (MVP)	Measured How
Meetings Booked	Number of qualified meetings from ABM outreach	4-8 per month	CRM deal creation linked to ABM sequence
Pipeline Value	Total ARR potential from ABM-sourced deals	Depends on deal size	CRM deal value where source = ABM
Reply Rate	% of contacts who reply (any sentiment)	> 8%	Replies / total contacts sequenced
Positive Reply Rate	% of contacts who reply positively	> 3%	Positive replies / total contacts sequenced
Signal-to-Meeting Conversion	% of P1 signals that ultimately result in a meeting	> 5%	Meetings / P1 signals ingested
Email Open Rate	% of emails opened	> 40%	Opens / emails delivered
LinkedIn Acceptance Rate	% of connection requests accepted	> 25%	Accepts / connection requests sent
AI QC Pass Rate	% of AI messages that pass QC on first attempt	> 70%	First-pass QC approvals / total generations
Human Override Rate	% of AI-approved messages that humans edit or reject	< 15%	Human edits + rejections / AI approvals

13.2 — Feedback Loops

Loop 1 — Signal Quality Feedback

- Every 30 days, analyze: Which signal types lead to meetings? Which lead nowhere?
- If `HIRING_SURGE` signals lead to 0 meetings but `REGULATORY` signals lead to 5 → increase `REGULATORY` signal weight in scoring, decrease `HIRING_SURGE` weight

Loop 2 — Prompt Optimization Feedback

- Every 30 days, analyze: Which AI message variations got the highest reply rates?
- Extract patterns (length, angle, CTA type) from top-performing messages
- Update prompt templates with these patterns

Loop 3 — Scoring Weight Feedback

- Every 90 days, analyze correlation between each scoring dimension and actual meetings

- Adjust weights per rules in SCORE-RECAL-001/002/003

Loop 4 — Persona Performance Feedback

- Every 30 days, analyze: Which persona types respond most? CTOs or CDOs? Directors or VPs?
- Prioritize outreach to higher-responding personas in future sequences

SECTION 14 — STATE MANAGEMENT —

COMPLETE STATE MACHINE

Every account and every contact has a state. States determine what actions are allowed, what workflows can run, and what transitions are possible. This is the central coordination mechanism that prevents duplicate outreach, workflow collisions, and inconsistent behavior.

14.1 — Account State Machine

State	Meaning	Allowed Transitions	Transition Trigger
WATCHING	Account exists in system, being monitored for signals	→ ENRICHING (score hits HOT)	Score crosses 75 or manual promotion
ENRICHING	Active enrichment in progress	→ HOT (enrichment complete) → WATCHING (enrichment fails 3x)	Enrichment workflow completion or failure
HOT	Enriched and ready for AI personalization + outreach	→ OUTREACH_ACTIVE (message approved + sent)	First message sent to any contact
OUTREACH_ACTIVE	Sequences are running	→ ENGAGED (positive reply) → HOT (all sequences complete, no reply)	Reply received or all sequences finish
ENGAGED	Positive response received, human involved	→ MEETING_BOOKED (meeting scheduled) → OUTREACH_ACTIVE (re-engagement after 45 days)	Meeting event or cooldown timer
MEETING_BOOKED	Qualified meeting scheduled	→ OPPORTUNITY (deal created in CRM) → ENGAGED (meeting no-show, reschedule)	CRM deal creation or no-show event
OPPORTUNITY	Active deal in pipeline	→ CUSTOMER (closed-won) → DORMANT (closed-lost)	CRM deal stage change
CUSTOMER	Existing Decimal client	→ OPPORTUNITY (upsell identified)	New upsell signal or manual trigger
DORMANT	Lost deal or disengaged	→ WATCHING (after 90-day cooldown + new P1 signal)	Timer expiry + new signal

CRITICAL: State transitions are **ONE-WAY** along the primary path. An account in **OUTREACH_ACTIVE** cannot skip to **MEETING_BOOKED** without going through **ENGAGED**. The only exception is **DORMANT** → **WATCHING** which is a reset.

14.2 — Contact State Machine

State	Meaning	Transitions
UNCONTACTED	No outreach sent yet	→ SEQUENCED (first message sent)
SEQUENCED	Active outreach sequence running	→ OPENED (first open) → REPLIED (reply received without open tracking)

State	Meaning	Transitions
OPENED	Contact opened at least one message	→ REPLIED (reply received)
REPLIED	Contact replied (any sentiment)	→ QUALIFIED (positive + human confirms) → DISQUALIFIED (negative + confirmed)
QUALIFIED	Confirmed interested contact	Terminal state for contact. Account-level tracking takes over.
DISQUALIFIED	Confirmed not interested / wrong person	No further outreach unless manual override by account_owner.

SECTION 15 — RETRY, FAILURE & RECOVERY LOGIC

Enterprise systems must assume that everything fails. APIs time out. AI models return errors. Emails bounce. Enrichment providers go down. The system must handle every failure gracefully.

15.1 — Retry Strategy by Failure Type

Failure Type	Retry Strategy	Max Retries	After Max Retries
API timeout (enrichment, CRM)	Exponential backoff: 30s, 1m, 5m, 15m	4	Dead-letter queue. Manual handling within 24h.
AI model failure (timeout/error)	Retry with same model after 60s. If fails, try fallback model.	2 (primary) + 1 (fallback)	Queue for manual message creation.
Email delivery failure	No retry for hard bounces. Soft bounce: retry once after 4 hours.	1 for soft bounce	Mark email as UNDELIVERABLE. Switch to LinkedIn.
LinkedIn API failure	Retry after 15 minutes.	3	Skip this LinkedIn touch. Continue to next step.
CRM sync failure	Exponential: 1m, 5m, 15m, 60m	4	Dead-letter queue. Critical — manual sync required.
Webhook failure (engagement)	Retry every 5 minutes for 1 hour	12	Log as missed event. Flag for manual review.

[RETRY-001]

Every retry attempt must be logged: failure_type, attempt_number, timestamp, error_message, result.

[RETRY-002]

Dead-letter queues must be reviewed by a human at least once per business day.

[RETRY-003]

If any single service has a failure rate above 20% in a 1-hour window → trigger system alert. Something is fundamentally wrong — don't keep retrying.

SECTION 16 — TIMING MATRIX — EVERY CLOCK IN THE SYSTEM

This is a single reference for every time-based rule in the entire ABM engine.

Timer	Duration	Context	What Happens When It Expires
Signal polling — News	Every 4 hours	Google News API	New poll cycle starts
Signal polling — SAMA	Every 6 hours	SAMA RSS	New poll cycle starts
Signal polling — LinkedIn	Every 12 hours	LinkedIn data	New poll cycle starts
Signal polling — Funding	Every 24 hours	Funding databases	New poll cycle starts
Signal freshness — FRESH	0-7 days	Signal age	Transitions to AGING
Signal freshness — AGING	8-21 days	Signal age	Transitions to STALE
Signal freshness — STALE	22-45 days	Signal age	Transitions to EXPIRED
Signal freshness — EXPIRED	46+ days	Signal age	Excluded from scoring
Enrichment timeout	30 seconds per API call	Enrichment API	Retry once, then proceed with available data
Email verification timeout	15 seconds	Verification API	Mark UNVERIFIED, retry next cycle
Email sequence — Touch 1 to 2	3 days	Sequence timer	Send Touch 2
Email sequence — Touch 2 to 3	4 days	Sequence timer	Send Touch 3
Email sequence — Touch 3 to 4	6 days	Sequence timer	Send Touch 4
LinkedIn — Connection wait	7 days	After connection request	Mark as 'connection_pending'
LinkedIn — Step 1 to 2	2 days after accept	After connection accepted	Send Value DM
LinkedIn — Step 2 to 3	7 days	After Value DM	Send follow-up
Cross-channel gap	72 hours	Email first → LinkedIn first touch	LinkedIn outreach can begin
Human review SLA	4 hours (business hours)	Message pending review	Escalate to backup reviewer
Handoff acknowledgment	48 hours	Sales team response	Escalate to sales manager
Sequence cooldown	45 days	After sequence completes	Contact eligible for new sequence

Timer	Duration	Context	What Happens When It Expires
HOT enrichment refresh	30 days	Since last enrichment	Trigger re-enrichment
WARM enrichment refresh	90 days	Since last enrichment	Trigger light enrichment
Dormant cooldown	90 days	After deal lost	Account eligible for re-engagement
Manual override expiry	30 days	After admin tier override	Scoring engine takes over
CRM sync poll	30 minutes	Deal stage changes	Check for updates
Scoring recalibration	90 days	System optimization	Re-evaluate dimension weights
Domain warmup — Week 1	7 days	New email domain	Max 5 emails/day
Domain warmup — Week 2	7 days	Email domain	Max 10 emails/day
Domain warmup — Week 3	7 days	Email domain	Max 15 emails/day

SECTION 17 — KSA-SPECIFIC BUSINESS RULES

The KSA banking ecosystem has unique cultural, regulatory, and operational characteristics that must be hardcoded into the ABM engine. These are not optional — they are structural requirements.

17.1 — Work Week & Business Hours

[KSA-001]

KSA business week: Sunday through Thursday. Friday and Saturday are the weekend.

[KSA-002]

Business hours: 9:00 AM — 6:00 PM AST (Arabia Standard Time, UTC+3).

[KSA-003]

NO outreach emails or LinkedIn messages should be sent on Friday or Saturday. The system must queue messages for the next business day (Sunday).

[KSA-004]

NO outreach during Ramadan (dates vary yearly — must be configured). During Ramadan, outreach volume is reduced to 50% and send times shift to 10:00 AM — 2:00 PM only.

[KSA-005]

NO outreach during Eid al-Fitr (3 days) or Eid al-Adha (4 days). Queue all messages for the first business day after Eid.

17.2 — Regulatory Compliance

[KSA-010]

SAMA CSF (Cyber Security Framework) compliance is mandatory for ANY vendor selling to Saudi banks. Every outreach message to a bank must be able to reference SAMA CSF alignment if asked.

[KSA-011]

KSA Data Residency: All data about Saudi bank contacts must be stored on infrastructure that complies with Saudi data localization requirements. The technical team must ensure this.

[KSA-012]

PDPL (Personal Data Protection Law): Outreach must comply with Saudi data protection regulations. All contacts have the right to request data deletion.

17.3 — Cultural Communication Rules

[KSA-020]

Relationship-first culture: The FIRST outreach to any Saudi executive must NOT be a hard sell. It must be value-led, consultative, and respectful of the relationship-building process.

[KSA-021]

Islamic banking: When outreaching to Islamic banks, messaging must acknowledge Sharia compliance requirements. Never reference interest-based products — use 'profit-sharing', 'Murabaha', 'Ijara' terminology where relevant.

[KSA-022]

Titles and honorifics: Use formal titles in all initial outreach. 'Dear Dr. Al-Furaih' not 'Hi Saleh'. Only switch to first names if the contact initiates informal communication.

[KSA-023]

Saudi Vision 2030 alignment: Many KSA banks are actively aligning with Vision 2030 goals (financial sector development, digital transformation, financial inclusion). Messaging should connect Decimal's capabilities to these national priorities where relevant.

SECTION 18 — EDGE CASES & EXCEPTION HANDLING

18.1 — Signal Edge Cases

[EDGE-SIG-001]

Signal about an account NOT in the system: Create a new account record with state WATCHING and tier COLD. Run the signal through normal processing. If the new account scores HOT immediately → follow standard HOT flow.

[EDGE-SIG-002]

Signal mentions MULTIPLE accounts (e.g., 'Bank A and Bank B announce joint digital initiative'): Create one signal record per account. Both get the same content but independent processing.

[EDGE-SIG-003]

Signal about a Decimal COMPETITOR winning a deal at target account: This is a NEGATIVE signal. Reduce relationship_warmth by 20 points on next score calculation. Change account_state to DORMANT with reason: 'Competitor won'. Cooldown: 180 days (double the standard 90).

[EDGE-SIG-004]

Signal is in Arabic: AI must translate to English for normalization. Store both Arabic original AND English translation. Confidence_score reduced by 0.1 if machine-translated (until human validates).

18.2 — Scoring Edge Cases

[EDGE-SCORE-001]

Account has a perfect score (100) but zero contacts: Tier is capped at WARM. WHY: Cannot execute outreach without contacts. The high score triggers priority enrichment to find contacts.

[EDGE-SCORE-002]

Account score fluctuates between 73-77 repeatedly (hovering at HOT/WARM boundary): Implement hysteresis band. Once HOT, must drop to 70 (not 74) to be downgraded to WARM. Once WARM, must reach 78 (not 75) to be upgraded to HOT. WHY: Prevents rapid tier flipping that would confuse workflows.

[EDGE-SCORE-003]

Two accounts at the same bank (e.g., SNB Group and SNB Digital Ventures): Treat as separate accounts with separate scores. But link them as 'related_accounts'. If one becomes HOT, flag the other for human review on whether to coordinate outreach.

18.3 — Outreach Edge Cases

[EDGE-OUT-001]

Contact leaves the company during active sequence: If enrichment detects role change → STOP all sequences for this contact. Mark as DISQUALIFIED with reason: 'Left company'. DO NOT auto-create a new contact at their new company — that's a separate account decision.

[EDGE-OUT-002]

Contact's email starts bouncing mid-sequence: After first bounce, wait 48 hours and retry ONCE. If second attempt bounces → mark email INVALID, switch remaining touches to LinkedIn-only. If no LinkedIn → mark contact as UNREACHABLE.

[EDGE-OUT-003]

Two contacts at the same account reply on the same day: Both replies trigger handoff. The handoff package shows BOTH replies. Account owner manages the coordination. System does NOT send any auto-response to either.

[EDGE-OUT-004]

Contact replies AFTER the sequence is complete (i.e., all 4 email touches were sent and the sequence closed): Still classify the reply. Still trigger handoff if positive. The sequence being complete does not mean the account is abandoned.

18.4 — System Edge Cases

[EDGE-SYS-001]

AI model is down for extended period (2+ hours): Queue all AI tasks. Do NOT fall back to template-based messaging. Wait for the AI to recover. WHY: Template messages violate the 'reasoning before messaging' principle. It's better to delay than to send generic content.

[EDGE-SYS-002]

Enrichment provider returns obviously wrong data (e.g., claims a bank has 5 employees): Flag data as SUSPICIOUS. Do NOT use it for scoring or outreach. Log for human review.

[EDGE-SYS-003]

Same person appears as a contact at two different accounts (e.g., board member at two banks): Allow the duplicate. They ARE two separate relationships. But coordinate outreach — never send to the same person from two different account sequences simultaneously. Minimum 14-day gap between touches across accounts.

SECTION 19 — GLOSSARY OF BUSINESS TERMS

Term	Definition
ABM	Account-Based Marketing — a strategy that treats each target organization as a market of one
Account	A target organization (bank, fintech, NBFC) that Decimal could sell to
Buying Committee	The group of people at an account who influence or make the purchasing decision
Signal	An external market event that indicates buying intent or strategic relevance
Tier	HOT / WARM / COLD classification based on account score
Sequence	A planned series of outreach touches to a specific contact
Persona	The role-based profile of a contact (CTO, CDO, VP Engineering, etc.)
Pain Inference	The AI process of deducing operational problems from signals and context
Handoff	The transition from AI-automated outreach to human-led sales conversation
Freshness	How recent a signal is — affects its impact on scoring
Enrichment	The process of gathering detailed intelligence about an account and its contacts
QC	Quality Control — validation of AI outputs before they reach prospects
State	The current lifecycle stage of an account or contact
Engagement	Any interaction a prospect has with our outreach (open, click, reply, accept)
Cooldown	A mandatory waiting period before re-engaging an account or contact
Dead-letter queue	A holding area for failed operations that need human intervention
Hysteresis band	A buffer zone around tier thresholds to prevent rapid tier flipping
SAMA	Saudi Arabian Monetary Authority — the central bank and financial regulator
PDPL	Personal Data Protection Law — Saudi Arabia's data protection regulation
ICP	Ideal Customer Profile — the characteristics of a perfect target account
SDR	Sales Development Representative — handles initial qualification
AE	Account Executive — handles deal negotiation and closure
CTA	Call-to-Action — the specific ask in an outreach message
P1/P2/P3/P4	Signal priority levels from Critical to Low
MVP	Minimum Viable Product — the first operational version of the system

END OF DOCUMENT

This document is the single source of truth for all ABM engine business logic.

If it is not specified here, it does not exist in the system.

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